Submission Summary

Conference Name
International Conference on Technology Driven Transformation for Smart , Secure and Sustainable Future (AVDHARAN-ICTDTS3F-2026)

Track Name
Smart and Intelligent Computing

Paper ID
60

Paper Title
Image-Based Breed Recognition of Indian Cattle and Buffaloes

Abstract
Accurate identification of Indian cattle and buffalo breeds is important for livestock management, dairy operations, and breed conservation. This paper proposes an image-based breed recognition system using Computer Vision and Deep Learning. The proposed two-stage approach uses YOLOv8 to detect and localise the animal, followed by a ResNet50-based transfer learning classifier to identify the breed from the extracted region of interest (ROI). The system is designed to reduce the influence of complex backgrounds and handle variations in animal appearance, lighting, and camera viewpoint. A Streamlit-based interface is proposed for image upload and prediction display. The system will be evaluated using accuracy, precision, recall, F1-score, and confusion matrix analysis. The proposed approach aims to support efficient and consistent preliminary breed identification.

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